

Duality

Lecture 5

October 6, 2026

Recap From Last Time

Primal-Dual Pair of Problems

	Primal ($\mathcal{P}$)	
minimize	$c^T x$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq}$
	$x_j \geq 0,$	$\forall j \in J_p$
	$x_j \leq 0,$	$\forall j \in J_n$
	x_j free,	$\forall j \in J_f$
variables	$x \in \mathbb{R}^n$	

Start with a primal minimization

Seek **lower bounds** on optimal value p^*

Recap From Last Time

Primal-Dual Pair of Problems

Primal ($\mathcal{P}$)		Dual ($\mathcal{D}$)	
minimize	$\overset{x}{c}^T x$	maximize	$\overset{\lambda}{\lambda}^T b$
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i, \quad \forall i \in I_{ge}$	$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i, \quad \forall i \in I_{le}$	$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i, \quad \forall i \in I_{eq}$	λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0, \quad \forall j \in J_p$	$(x_j \rightarrow) \quad \lambda^T A_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0, \quad \forall j \in J_n$	$(x_j \rightarrow) \quad \lambda^T A_j \geq c_j,$	$\forall j \in J_n$
	x_j free, $\forall j \in J_f$	$(x_j \rightarrow) \quad \lambda^T A_j = c_j,$	$\forall j \in J_f$
variables	$x \in \mathbb{R}^n$	variables	$\lambda \in \mathbb{R}^m.$

Start with a primal minimization

Seek **lower bounds** on optimal value p^*

Construct the dual problem

Talked about a general recipe to do take dual of **any** primal

Weak duality

Primal ($\mathcal{P}$)

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & \mathbf{c}^T \mathbf{x} \\ (\lambda_i \rightarrow) \quad & \mathbf{a}_i^T \mathbf{x} \geq b_i, \quad \forall i \in I_{\text{ge}}, \\ (\lambda_i \rightarrow) \quad & \mathbf{a}_i^T \mathbf{x} \leq b_i, \quad \forall i \in I_{\text{le}}, \\ (\lambda_i \rightarrow) \quad & \mathbf{a}_i^T \mathbf{x} = b_i, \quad \forall i \in I_{\text{eq}}, \\ & \mathbf{x}_j \geq 0, \quad \forall j \in J_p, \\ & \mathbf{x}_j \leq 0, \quad \forall j \in J_n, \\ & \mathbf{x}_j \text{ free}, \quad \forall j \in J_f. \end{aligned}$$

Dual ($\mathcal{D}$)

$$\begin{aligned} \text{maximize}_{\lambda} \quad & \lambda^T \mathbf{b} \\ & \lambda_i \geq 0, \quad \forall i \in I_{\text{ge}}, \\ & \lambda_i \leq 0, \quad \forall i \in I_{\text{le}}, \\ & \lambda_i \text{ free}, \quad \forall i \in I_{\text{eq}}, \\ (\mathbf{x}_j \rightarrow) \quad & \lambda^T \mathbf{A}_j \leq c_j, \quad \forall j \in J_p, \\ (\mathbf{x}_j \rightarrow) \quad & \lambda^T \mathbf{A}_j \geq c_j, \quad \forall j \in J_n, \\ (\mathbf{x}_j \rightarrow) \quad & \lambda^T \mathbf{A}_j = c_j, \quad \forall j \in J_f. \end{aligned}$$

Weak duality

Primal ($\mathcal{P}$)			Dual ($\mathcal{D}$)		
minimize _{x}	$c^T x$		maximize _{λ}	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge},$	$\lambda_i \geq 0,$		$\forall i \in I_{ge},$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le},$	$\lambda_i \leq 0,$		$\forall i \in I_{le},$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq},$	λ_i free,		$\forall i \in I_{eq},$
	$x_j \geq 0,$	$\forall j \in J_p,$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p,$
	$x_j \leq 0,$	$\forall j \in J_n,$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n,$
	x_j free,	$\forall j \in J_f.$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f.$

Theorem (Weak duality)

If x is feasible for ($\mathcal{P}$) and λ is feasible for ($\mathcal{D}$), then $\lambda^T b \leq c^T x$.

Proof. Trivially true from our construction – omitted.

Weak duality

Primal ($\mathcal{P}$)			Dual ($\mathcal{D}$)		
minimize _{x}	$c^T x$		maximize _{λ}	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge},$	$\lambda_i \geq 0,$		$\forall i \in I_{ge},$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le},$	$\lambda_i \leq 0,$		$\forall i \in I_{le},$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq},$	λ_i free,		$\forall i \in I_{eq},$
	$x_j \geq 0,$	$\forall j \in J_p,$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p,$
	$x_j \leq 0,$	$\forall j \in J_n,$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n,$
	x_j free,	$\forall j \in J_f.$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f.$

Theorem (Weak duality)

If x is feasible for ($\mathcal{P}$) and λ is feasible for ($\mathcal{D}$), then $\lambda^T b \leq c^T x$.

Proof. Trivially true from our construction – omitted.

Discussed a few simple implications ...

How Good Is This Dual (Lower) Bound?

Primal ($\mathcal{P}$)			Dual ($\mathcal{D}$)		
minimize _{x}	$c^T x$		maximize _{λ}	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge},$	$\lambda_i \geq 0,$		$\forall i \in I_{ge},$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le},$	$\lambda_i \leq 0,$		$\forall i \in I_{le},$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq},$	λ_i free,		$\forall i \in I_{eq},$
	$x_j \geq 0,$	$\forall j \in J_p,$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p,$
	$x_j \leq 0,$	$\forall j \in J_n,$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n,$
	x_j free,	$\forall j \in J_f.$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f.$

How Good Is This Dual (Lower) Bound?

Primal ($\mathcal{P}$)			Dual ($\mathcal{D}$)		
minimize _{x}	$c^T x$		maximize _{λ}	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge},$	$\lambda_i \geq 0,$		$\forall i \in I_{ge},$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le},$	$\lambda_i \leq 0,$		$\forall i \in I_{le},$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq},$	λ_i free,		$\forall i \in I_{eq},$
	$x_j \geq 0,$	$\forall j \in J_p,$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p,$
	$x_j \leq 0,$	$\forall j \in J_n,$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n,$
	x_j free,	$\forall j \in J_f.$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f.$

Theorem (Strong duality)

If $(\mathcal{P})$ has an optimal solution, so does $(\mathcal{D})$, and the optimal values are equal, $\lambda^ = d^*$.*

How Good Is This Dual (Lower) Bound?

Primal ($\mathcal{P}$)			Dual ($\mathcal{D}$)		
minimize _{x}	$c^T x$		maximize _{λ}	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge},$	$\lambda_i \geq 0,$		$\forall i \in I_{ge},$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le},$	$\lambda_i \leq 0,$		$\forall i \in I_{le},$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq},$	λ_i free,		$\forall i \in I_{eq},$
	$x_j \geq 0,$	$\forall j \in J_p,$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p,$
	$x_j \leq 0,$	$\forall j \in J_n,$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n,$
	x_j free,	$\forall j \in J_f.$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f.$

Theorem (Strong duality)

If $(\mathcal{P})$ has an optimal solution, so does $(\mathcal{D})$, and the optimal values are equal, $\lambda^ = d^*$.*

Proof. Many proofs possible; Bertsimas & Tsitsiklis prove using simplex algorithm

- We provide a more general proof, in three steps:
 1. The **separating hyperplane theorem** (for convex sets)
 2. The Farkas Lemma
 3. Strong duality
- Need a tiny bit of **real analysis** background...

A Few Real Analysis Results

Definition (Closed Set)

A set $S \subseteq \mathbb{R}^n$ is called **closed** if it contains the limit of any sequence of elements of S . That is, if $x_n \in S$, $\forall n \geq 1$ and $x_n \rightarrow x^*$, then $x^* \in S$.

A Few Real Analysis Results

Definition (Closed Set)

A set $S \subseteq \mathbb{R}^n$ is called **closed** if it contains the limit of any sequence of elements of S . That is, if $x_n \in S$, $\forall n \geq 1$ and $x_n \rightarrow x^*$, then $x^* \in S$.

Theorem

Every polyhedron is closed.

A Few Real Analysis Results

Definition (Closed Set)

A set $S \subseteq \mathbb{R}^n$ is called **closed** if it contains the limit of any sequence of elements of S . That is, if $x_n \in S$, $\forall n \geq 1$ and $x_n \rightarrow x^*$, then $x^* \in S$.

Theorem

Every polyhedron is closed.

Proof.

- Consider $P = \{x \in \mathbb{R}^n \mid Ax \geq b\}$ (representation is w.l.o.g.)
- Suppose that $\{x_n\}_{n \geq 1}$ is a sequence with $x_n \in P$ for every n , and $x_n \rightarrow x^*$.
- For each k , we have $x_k \in P$, and therefore, $Ax_k \geq b$.
- Then, $Ax^* = A(\lim_{k \rightarrow \infty} x_k) = \lim_{k \rightarrow \infty} Ax_k \geq b$, so x^* belongs to P . □

A Few Real Analysis Results

Definition (Closed Set)

A set $S \subseteq \mathbb{R}^n$ is called **closed** if it contains the limit of any sequence of elements of S . That is, if $x_n \in S$, $\forall n \geq 1$ and $x_n \rightarrow x^*$, then $x^* \in S$.

Theorem

Every polyhedron is closed.

*Is every **convex set** closed?*

A Few Real Analysis Results

Definition (Closed Set)

A set $S \subseteq \mathbb{R}^n$ is called **closed** if it contains the limit of any sequence of elements of S . That is, if $x_n \in S$, $\forall n \geq 1$ and $x_n \rightarrow x^*$, then $x^* \in S$.

Theorem

Every polyhedron is closed.

Theorem (Weierstrass' Theorem)

*If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous function, and if S is a **nonempty, closed, and bounded** subset of $\mathbb{R}^n$, then there exist $\underline{x}, \bar{x} \in S$ such that $f(\underline{x}) \leq f(x) \leq f(\bar{x})$ for all $x \in S$.*

i.e., a continuous function achieves its minimum and maximum

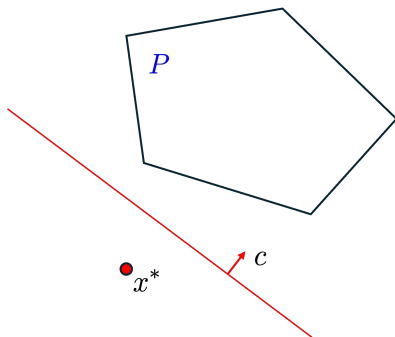
Separating Hyperplane Theorem

The first **fundamental result in optimization**

Separating Hyperplane Theorem

Theorem (**Simple** Separating Hyperplane Theorem)

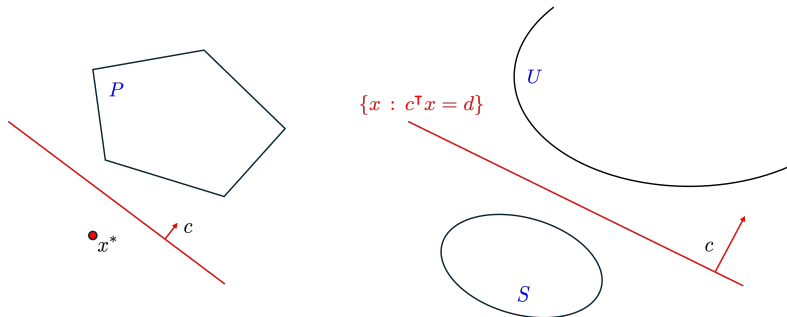
Consider a point x^* and a polyhedron P . If $x^* \notin P$, then there exists a vector $c \in \mathbb{R}^n$ such that $c \neq 0$ and $c^T x^* < c^T y$ holds for all $y \in P$.



Separating Hyperplane Theorem

Theorem (Separating Hyperplane Theorem **for Convex Sets**)

Let S and U be two nonempty, closed, convex subsets of $\mathbb{R}^n$ such that $S \cap U = \emptyset$ and S is bounded. Then, there exists $c \in \mathbb{R}^n$ and $d \in \mathbb{R}$ such that $S \subset \{x \in \mathbb{R}^n : c^T x < d\}$ and $U \subset \{x \in \mathbb{R}^n : c^T x > d\}$.



Separating Hyperplane Theorem

Theorem (Separating Hyperplane Theorem **for Convex Sets**)

Let S and U be two nonempty, closed, convex subsets of $\mathbb{R}^n$ such that $S \cap U = \emptyset$ and S is bounded. Then, there exists $c \in \mathbb{R}^n$ and $d \in \mathbb{R}$ such that $S \subset \{x \in \mathbb{R}^n : c^T x < d\}$ and $U \subset \{x \in \mathbb{R}^n : c^T x > d\}$.

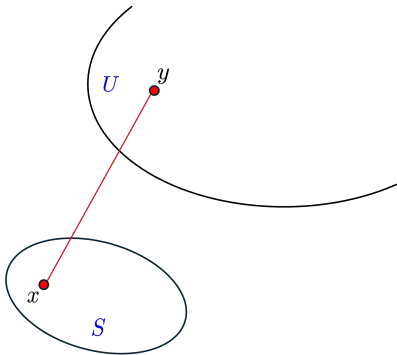
Proof.

Separating Hyperplane Theorem

Theorem (Separating Hyperplane Theorem for Convex Sets)

Let S and U be two nonempty, closed, convex subsets of $\mathbb{R}^n$ such that $S \cap U = \emptyset$ and S is bounded. Then, there exists $c \in \mathbb{R}^n$ and $d \in \mathbb{R}$ such that $S \subset \{x \in \mathbb{R}^n : c^T x < d\}$ and $U \subset \{x \in \mathbb{R}^n : c^T x > d\}$.

Proof. Consider $\|x - y\|$ with $x \in S, y \in U$

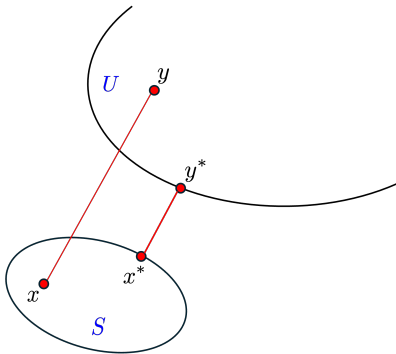


Separating Hyperplane Theorem

Theorem (Separating Hyperplane Theorem for Convex Sets)

Let S and U be two nonempty, closed, convex subsets of $\mathbb{R}^n$ such that $S \cap U = \emptyset$ and S is bounded. Then, there exists $c \in \mathbb{R}^n$ and $d \in \mathbb{R}$ such that $S \subset \{x \in \mathbb{R}^n : c^T x < d\}$ and $U \subset \{x \in \mathbb{R}^n : c^T x > d\}$.

Proof. Argue that the minimum is achieved, at x^*, y^*

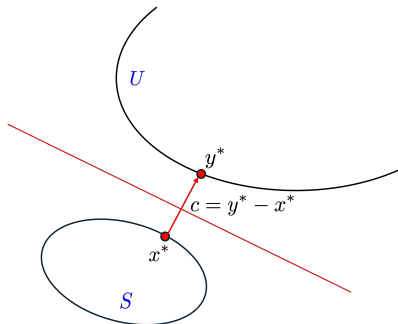


Separating Hyperplane Theorem

Theorem (Separating Hyperplane Theorem for Convex Sets)

Let S and U be two nonempty, closed, convex subsets of $\mathbb{R}^n$ such that $S \cap U = \emptyset$ and S is bounded. Then, there exists $c \in \mathbb{R}^n$ and $d \in \mathbb{R}$ such that $S \subset \{x \in \mathbb{R}^n : c^T x < d\}$ and $U \subset \{x \in \mathbb{R}^n : c^T x > d\}$.

Proof. Argue that $c = y^* - x^*$ and $d = \frac{c^T(x^* + y^*)}{2}$ give strict separating hyperplane



Separating Hyperplane Theorem - Caveats!

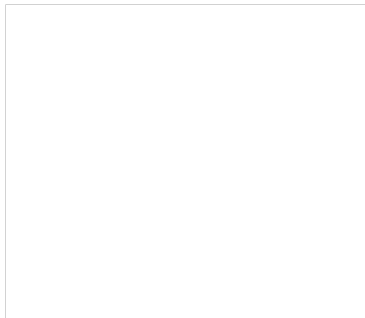
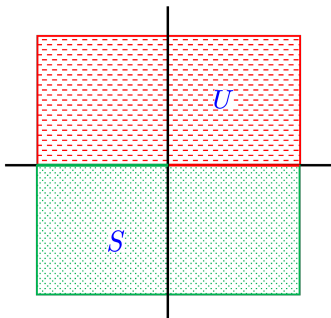
Both conditions in the theorem needed: **closed** and at least one set **bounded**

Separating Hyperplane Theorem - Caveats!

Both conditions in the theorem needed: **closed** and at least one set **bounded**

- **Left:** two convex sets that are **not closed** but are both bounded:

$$S = [-1, 1] \times [-1, 0) \cup \{(x, y) : x \in [-1, 0], y = 0\}, \quad U = [-1, 1]^2 \setminus S$$



Separating Hyperplane Theorem - Caveats!

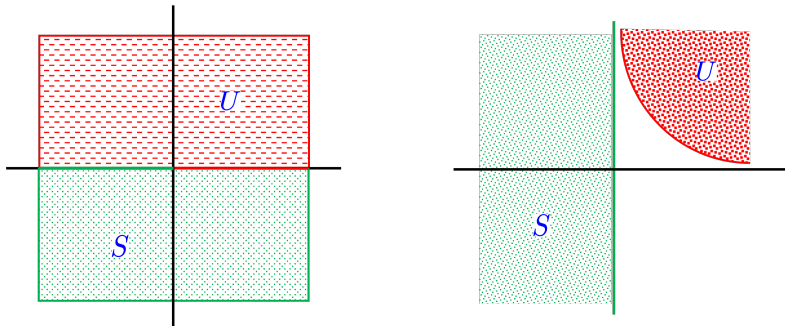
Both conditions in the theorem needed: **closed** and at least one set **bounded**

- **Left:** two convex sets that are **not closed** but are both bounded:

$$S = [-1, 1] \times [-1, 0) \cup \{(x, y) : x \in [-1, 0], y = 0\}, \quad U = [-1, 1]^2 \setminus S$$

- **Right:** two convex sets that are both closed but are **unbounded**

$$S = \{(x, y) : x \leq 0\}, \quad U = \{(x, y) : x \geq 0, y \geq 1/x\}$$

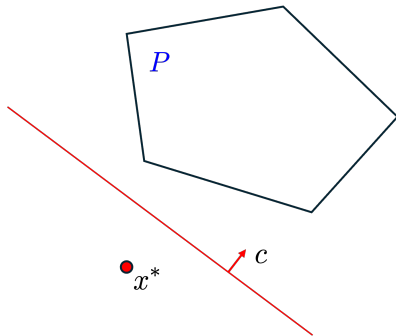


Needed For Our Purposes

We proved the first **fundamental result in optimization**!

Corollary (Needed for our purposes...)

If P is a polyhedron and $x^ \notin P$, there exists a hyperplane that strictly separates x^* from P , i.e., $\exists c \neq 0$ such that $c^T x^* < c^T x$ for any $x \in P$.*



Farkas Lemma

Time for the **second fundamental result in optimization!**

Farkas Lemma

Theorem (Farkas' Lemma)

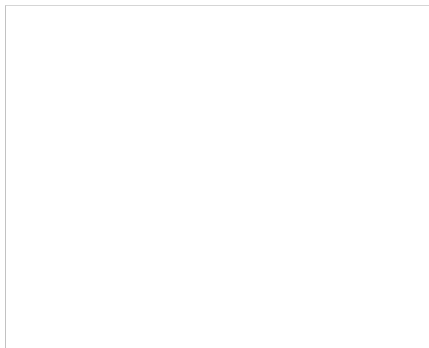
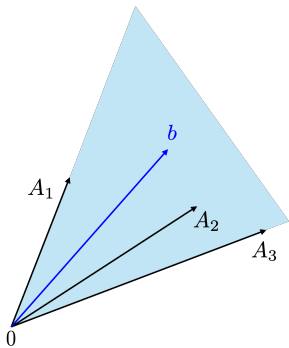
For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

(a) There exists some $x \geq 0$ such that $Ax = b$.

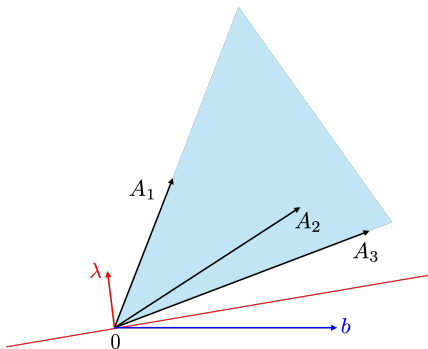
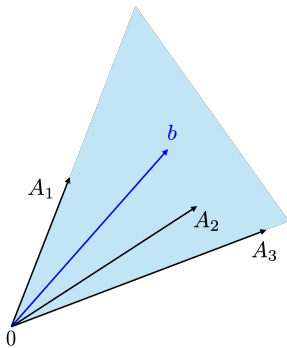


Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.



Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

(a) There exists some $x \geq 0$ such that $Ax = b$.

(b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

Proof. “(a) true implies (b) false.”

Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

(a) There exists some $x \geq 0$ such that $Ax = b$.

(b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

Proof. “(a) true implies (b) false.”

(a) true means $\exists x \geq 0 : Ax = b$.

(b) true means $\exists \lambda : \lambda^T A \geq 0$ and $\lambda^T b < 0$.

If (a) and (b) both true, then $\lambda^T b = \lambda^T Ax \geq 0$, which is a contradiction.

Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

(a) There exists some $x \geq 0$ such that $Ax = b$.

(b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

“(a) false implies (b) true.” Want to use the separating hyperplane theorem.

Farkas Lemma

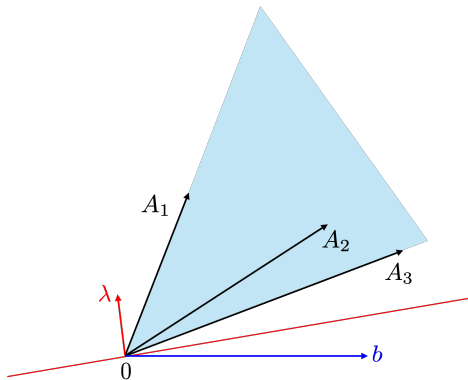
Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

“(a) false implies (b) true.” Want to use the separating hyperplane theorem.

- (a) false implies that $b \notin \{y : \exists x \geq 0 \text{ such that } y = Ax\} := S$.



Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

“(a) false implies (b) true.” Want to use the separating hyperplane theorem.

- (a) false implies that $b \notin \{y : \exists x \geq 0 \text{ such that } y = Ax\} := S$.
- S is a convex and **closed** set

Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

“(a) false implies (b) true.” Want to use the separating hyperplane theorem.

- (a) false implies that $b \notin \{y : \exists x \geq 0 \text{ such that } y = Ax\} := S$.
- S is a convex and **closed** set (S is polyhedral)

Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

“(a) false implies (b) true.” Want to use the separating hyperplane theorem.

- (a) false implies that $b \notin \{y : \exists x \geq 0 \text{ such that } y = Ax\} := S$.
- S is a convex and **closed** set (S is polyhedral)
- Separating Hyperplane Theorem implies $\exists \lambda : \lambda^T b < \lambda^T y, \forall y \in S$

Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

“(a) false implies (b) true.” Want to use the separating hyperplane theorem.

- (a) false implies that $b \notin \{y : \exists x \geq 0 \text{ such that } y = Ax\} := S$.
- S is a convex and **closed** set (S is polyhedral)
- Separating Hyperplane Theorem implies $\exists \lambda : \lambda^T b < \lambda^T y, \forall y \in S$
- $0 \in S \Rightarrow \lambda^T b < 0$

Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

“(a) false implies (b) true.” Want to use the separating hyperplane theorem.

- (a) false implies that $b \notin \{y : \exists x \geq 0 \text{ such that } y = Ax\} := S$.
- S is a convex and **closed** set (S is polyhedral)
- Separating Hyperplane Theorem implies $\exists \lambda : \lambda^T b < \lambda^T y, \forall y \in S$
- $0 \in S \Rightarrow \lambda^T b < 0$
- Every column A_i of A satisfies $\theta A_i \in S$ for every $\theta > 0$, so

$$\frac{\lambda^T b}{\theta} < \lambda^T A_i, \forall \theta > 0$$

Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

“(a) false implies (b) true.” Want to use the separating hyperplane theorem.

- (a) false implies that $b \notin \{y : \exists x \geq 0 \text{ such that } y = Ax\} := S$.
- S is a convex and **closed** set (S is polyhedral)
- Separating Hyperplane Theorem implies $\exists \lambda : \lambda^T b < \lambda^T y, \forall y \in S$
- $0 \in S \Rightarrow \lambda^T b < 0$
- Every column A_i of A satisfies $\theta A_i \in S$ for every $\theta > 0$, so

$$\frac{\lambda^T b}{\theta} < \lambda^T A_i, \forall \theta > 0$$

- Limit $\theta \rightarrow \infty$ implies $\lambda^T A_i \geq 0$. ■

Farkas Lemma Implications

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.*
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.*

We proved the **second fundamental result in optimization!**

Farkas Lemma Implications

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

We proved the **second fundamental result in optimization!**

- Suppose your primal problem $(\mathcal{P})$ was the standard-form LP:

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize} \quad c^T x \\ & \text{subject to} \quad Ax = b \\ & \quad \quad \quad x \geq 0 \end{array}$$

- What does the Farkas Lemma state about this?

Strong Duality

Consider the following primal-dual pair:

$$\begin{aligned} (\mathcal{P}) \quad & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \end{aligned}$$

Strong Duality

Consider the following primal-dual pair:

$$\begin{aligned} (\mathcal{P}) \quad & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \end{aligned}$$

Theorem (**Strong Duality**)

If $(\mathcal{P})$ has an optimal solution, so does $(\mathcal{D})$, and their optimal values are equal.

Strong Duality

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \\ (\mathcal{D}) & \text{maximize } \lambda^T b \\ & \text{subject to } \lambda^T A = c^T, \lambda \geq 0. \end{array}$$

Proof.

- Assume $(\mathcal{P})$ has optimal solution x^*
- Will prove that $(\mathcal{D})$ admits feasible solution λ such that $\lambda^T b = c^T x^*$

Strong Duality

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \end{array} \qquad \begin{array}{ll} (\mathcal{D}) & \text{maximize } \lambda^T b \\ & \text{subject to } \lambda^T A = c^T, \lambda \geq 0. \end{array}$$

Proof.

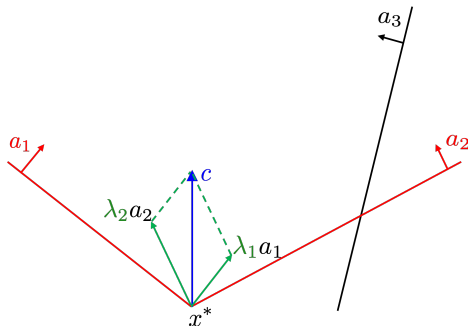
- Assume $(\mathcal{P})$ has optimal solution x^*
- Will prove that $(\mathcal{D})$ admits feasible solution λ such that $\lambda^T b = c^T x^*$
- Let $\mathcal{F} = \{i \mid a_i^T x^* = b_i\}$ denote the indices of active constraints at x^*
- Show that c can be written as conic combination of constraints $\{a_i : i \in \mathcal{F}\}$

Strong Duality

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \end{array} \quad \begin{array}{ll} (\mathcal{D}) & \text{maximize } \lambda^T b \\ & \text{subject to } \lambda^T A = c^T, \lambda \geq 0. \end{array}$$

Proof.

- Assume $(\mathcal{P})$ has optimal solution x^*
- Will prove that $(\mathcal{D})$ admits feasible solution λ such that $\lambda^T b = c^T x^*$
- Let $\mathcal{F} = \{i \mid a_i^T x^* = b_i\}$ denote the indices of active constraints at x^*
- Show that c can be written as conic combination of constraints $\{a_i : i \in \mathcal{F}\}$



Strong Duality

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \\ (\mathcal{D}) & \text{maximize } \lambda^T b \\ & \text{subject to } \lambda^T A = c^T, \quad \lambda \geq 0. \end{array}$$

Proof.

- First, we show that for any vector d , the following implication holds:

$$a_i^T d \geq 0, \forall i \in \mathcal{F} \Rightarrow c^T d \geq 0.$$

Strong Duality

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \\ (\mathcal{D}) & \text{maximize } \lambda^T b \\ & \text{subject to } \lambda^T A = c^T, \lambda \geq 0. \end{array}$$

Proof.

- First, we show that for any vector d , the following implication holds:

$$a_i^T d \geq 0, \forall i \in \mathcal{F} \Rightarrow c^T d \geq 0.$$

- For any such d , we claim that $x^* + \epsilon d \in P$ for small ϵ

Strong Duality

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \\ (\mathcal{D}) & \text{maximize } \lambda^T b \\ & \text{subject to } \lambda^T A = c^T, \lambda \geq 0. \end{array}$$

Proof.

- First, we show that for any vector d , the following implication holds:

$$a_i^T d \geq 0, \forall i \in \mathcal{F} \Rightarrow c^T d \geq 0.$$

- For any such d , we claim that $x^* + \epsilon d \in P$ for small ϵ
 - $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \in \mathcal{F}$ **for any** $\epsilon > 0$
 - $a_i^T x^* > b_i \forall i \notin \mathcal{F}$ implies that $\exists \epsilon > 0$ such that $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \notin \mathcal{F}$

Strong Duality

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \\ (\mathcal{D}) & \text{maximize } \lambda^T b \\ & \text{subject to } \lambda^T A = c^T, \lambda \geq 0. \end{array}$$

Proof.

- First, we show that for any vector d , the following implication holds:

$$a_i^T d \geq 0, \forall i \in \mathcal{F} \Rightarrow c^T d \geq 0.$$

- For any such d , we claim that $x^* + \epsilon d \in P$ for small ϵ
 - $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \in \mathcal{F}$ **for any** $\epsilon > 0$
 - $a_i^T x^* > b_i \forall i \notin \mathcal{F}$ implies that $\exists \epsilon > 0$ such that $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \notin \mathcal{F}$
- $c^T d \geq 0$ because otherwise $c^T(x^* + \epsilon d) < c^T x^*$ would contradict x^* optimal

Strong Duality

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \\ (\mathcal{D}) & \text{maximize } \lambda^T b \\ & \text{subject to } \lambda^T A = c^T, \lambda \geq 0. \end{array}$$

Proof.

- First, we show that for any vector d , the following implication holds:

$$a_i^T d \geq 0, \forall i \in \mathcal{F} \Rightarrow c^T d \geq 0.$$

- For any such d , we claim that $x^* + \epsilon d \in P$ for small ϵ
 - $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \in \mathcal{F}$ **for any** $\epsilon > 0$
 - $a_i^T x^* > b_i \forall i \notin \mathcal{F}$ implies that $\exists \epsilon > 0$ such that $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \notin \mathcal{F}$
- $c^T d \geq 0$ because otherwise $c^T(x^* + \epsilon d) < c^T x^*$ would contradict x^* optimal
- So $\nexists d : a_i^T d \geq 0, \forall i \in \mathcal{F}, c^T d < 0$

Strong Duality

$$\begin{array}{ll} (\mathcal{P}) & \text{minimize } c^T x \\ & \text{subject to } Ax \geq b \\ (\mathcal{D}) & \text{maximize } \lambda^T b \\ & \text{subject to } \lambda^T A = c^T, \lambda \geq 0. \end{array}$$

Proof.

- First, we show that for any vector d , the following implication holds:

$$a_i^T d \geq 0, \forall i \in \mathcal{F} \Rightarrow c^T d \geq 0.$$

- For any such d , we claim that $x^* + \epsilon d \in P$ for small ϵ
 - $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \in \mathcal{F}$ **for any** $\epsilon > 0$
 - $a_i^T x^* > b_i \forall i \notin \mathcal{F}$ implies that $\exists \epsilon > 0$ such that $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \notin \mathcal{F}$
- $c^T d \geq 0$ because otherwise $c^T(x^* + \epsilon d) < c^T x^*$ would contradict x^* optimal
- So $\nexists d : a_i^T d \geq 0, \forall i \in \mathcal{F}, c^T d < 0$
- Farkas Lemma : alternative (b) is not true, so alternative (a) must be true:

$$\exists \{\lambda_i\}_{i \in \mathcal{F}} : \lambda_i \geq 0, c = \sum_{i \in \mathcal{F}} \lambda_i a_i$$

Strong Duality

$$\begin{array}{ll}
 (\mathcal{P}) & \text{minimize } c^T x \\
 & \text{subject to } Ax \geq b \\
 (\mathcal{D}) & \text{maximize } \lambda^T b \\
 & \text{subject to } \lambda^T A = c^T, \quad \lambda \geq 0.
 \end{array}$$

Proof.

- First, we show that for any vector d , the following implication holds:

$$a_i^T d \geq 0, \forall i \in \mathcal{F} \Rightarrow c^T d \geq 0.$$

- For any such d , we claim that $x^* + \epsilon d \in P$ for small ϵ
 - $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \in \mathcal{F}$ **for any** $\epsilon > 0$
 - $a_i^T x^* > b_i \forall i \notin \mathcal{F}$ implies that $\exists \epsilon > 0$ such that $a_i^T(x^* + \epsilon d) \geq b_i, \forall i \notin \mathcal{F}$
- $c^T d \geq 0$ because otherwise $c^T(x^* + \epsilon d) < c^T x^*$ would contradict x^* optimal
- So $\nexists d : a_i^T d \geq 0, \forall i \in \mathcal{F}, c^T d < 0$
- Farkas Lemma : alternative (b) is not true, so alternative (a) must be true:

$$\exists \{\lambda_i\}_{i \in \mathcal{F}} : \lambda_i \geq 0, \quad c = \sum_{i \in \mathcal{F}} \lambda_i a_i$$

- Let $\lambda_i = 0$ for $i \notin \mathcal{F} \Rightarrow \exists \lambda$ feasible for $(\mathcal{D})$
- $\lambda^T b = \sum_{i \in \mathcal{F}} \lambda_i b_i = \sum_{i \in \mathcal{F}} \lambda_i a_i^T x^* = c^T x^*$ ■